2020 Paper E2.1: Digital Electronics II

Answer ALL questions.

There are THREE questions on the paper.

Question ONE counts for 50% of the marks, other questions 25% each

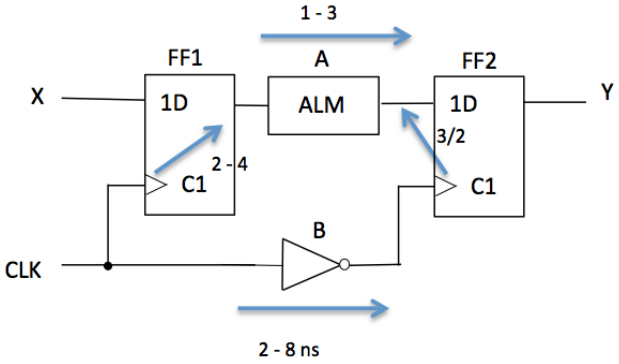
Time allowed: 2 hours

SOLUTIONS

(For Examiners Only)

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1. (a) This tests student's understanding of timing constraints in digital circuits.



(i)

$$\text{Setup:} \quad t_{cQ} + t_A + t_S \le T/2 + t_B$$

$$\text{Hold:} \quad t_B + t_H \le T/2 + t_{cQ} + t_A$$

[5]

(ii) First consider setup time constraints, we get:

$$t_{Pmax} + t_{Amax} + t_S - t_{Bmin} \le T_{min}/2$$

Therefore, $4 + 3 + 3 - 2 \le T_{min}/2$, $T_{min} \ge 16$ns.

Check hold time constraints:

$$t_{Bmax} + t_H \le \frac{T}{2} + t_{Pmin} + t_{Amin}$$

$$8 + 2 < 8 + 3 + 1.$$

Hold time constraint is ok.

Hence $f_{max} = 62.5$MHz.

[5]

(b) This question tests students understanding of address decoding and organisation of memory.

(i) 16kbyte require 14 bits of address bits. There address range is A[13:0].

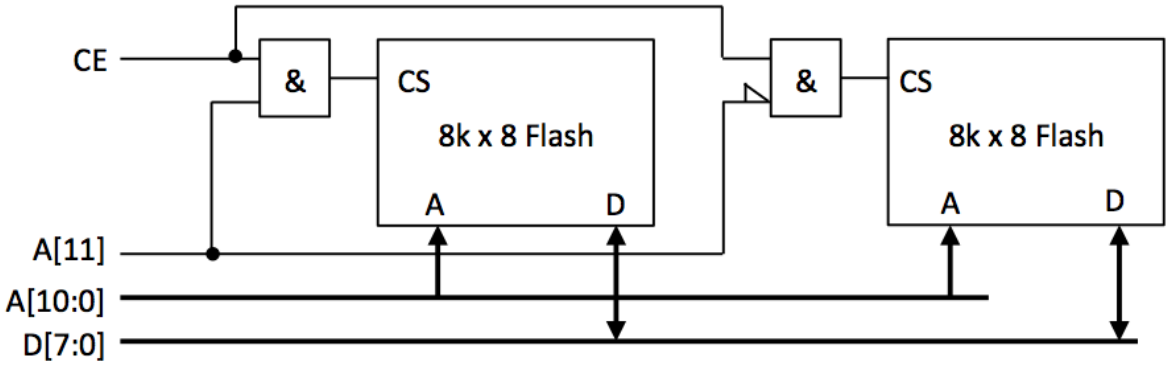
[2]

(ii) CE = RD & CLOCK & A15

$$CE = RD \;\&\; CLK \;\&\; A15 \;\&\; \overline{A14}$$

[4]

(iii)

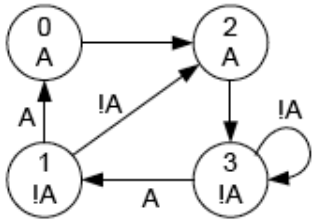


[4]

(c) This question tests student's ability to reverse engineering a simple FSM from the circuit, and to write FSM description in Verilog.

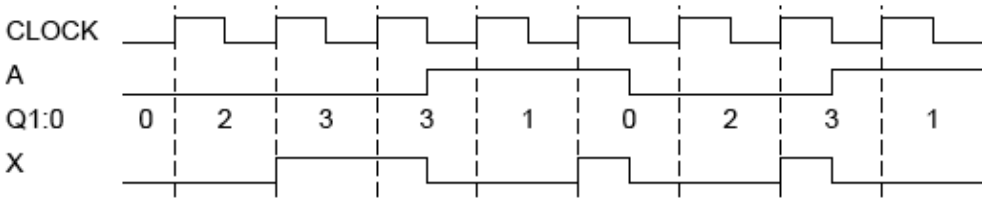
(i)

	D1:0/X	A=0	A=1
Q1:0	00	10/0	10/1
	01	10/1	00/0
	10	11/0	11/1
	11	11/1	01/0



[3]

(ii)



[3]

(iii)

```
module fsm (clk, a, x);

    input   a, clk;
    output  x;

    reg [1:0]   q;
    parameter   S0 = 2'd0, S1 = 2'd1,  S2 = 2'd2, S3 = 2'd3;

    initial state = S0;

    always @ (posedge clk)
        case (q)
            S0:  q <= S2;
            S1:  if (a == 1'b1) q <= S0;
                    else q <= S2;
            S2:  q <= S3;
            S3:  if (a == 1'b1) q <= S1;
                    else q <= S3;
            default: q <= S0;
        endcase

    always @ (*)
        case (q)
            S0: x = a;
            S1: x = a;
            S2: x = !a;
            S3: x = !a;
            default: x = a;
        endcase
endmodule
```

[4]

(d) This tests student's understanding of linear feedback registers and how to specify these in Verilog..

(i) & (ii)

Cycle	S	Q5	Q4	Q3	Q2	Q1
1	0	1	1	1	1	1
2	0	1	1	1	1	0
3	0	1	1	1	0	1
4	0	1	1	0	1	0
5	0	1	0	1	0	1
6	1	0	1	0	1	1
7	1	1	0	1	1	0
8	1	0	1	1	0	0
9	1	1	1	0	0	1
10	1	1	0	0	1	1

[3] & [3]

(iii)

```
module prbs (clk, s, q);

    input               clk;
    input               s;
    output [5:1]        q;

    reg [5:1]       sreg;

    initial sreg = 5'b11111;

    always @ (posedge clk)
        if (s==1'b0)    sreg <= {sreg[4:1], sreg[4] ^ sreg[1]};
        else            sreg <= {sreg[4:1], sreg[5] ^ sreg[3]};

    assign q = sreg;
endmodule
```

[4]

(e) This tests student's understanding of the basic principle of R-2D network in a DAC converter circuit.

(i) The two terminals on the right of the switch always connect to zero potential. In the down position, the switch is connected to ground. In the up position, the switch is connected to the virtual earth of the inverting opamp. Therefore the left terminal of the switch is always connected to zero volt.

[3]

(ii) The effective resistance at each node of the R-2R ladder network is 10kΩ. Therefore the total current from the voltage source to the ladder network is 2V/10k = 0.2mA.

This current is halved at each node.

Hence I2 = 0.1mA, I1 = 0.05mA, I0 = 0.025mA.

[3]

(iii) When X[2:0] = 5, the current flowing into the virtual earth of the opamp is 0.125mA. Therefore VOUT = -0.125 x 20 V = - 2.5V. The negative sign is because the opamp is an inverting amplifier.

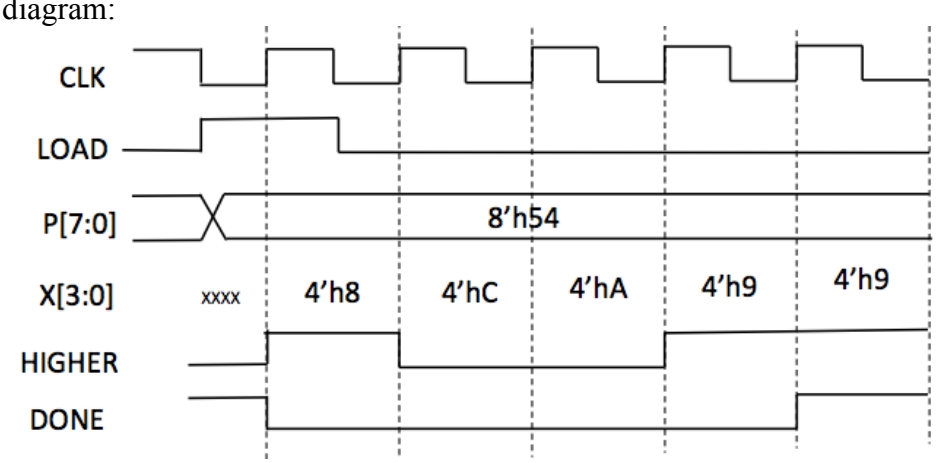
[4]

2. This question tests student's understanding the principle of successive approximation algorithm and how to apply it..

(a) Students may first draw this table to help, but not required.

X[3:0]	X^2	HIGHER	RESULT
4h8	8h40	TRUE	X[3] = 1
4hC	8h90	FALSE	X[2]= 0
4hA	8h64	FALSE	X[1] = 0
4h9	8h51	TRUE	X[0] = 1

Timing diagram:



[9]

(b) The principle of SAR is really bookwork. However, the application here is rather special. Instead of the DAC in a SAR based ADC, we use the squaring circuit. The rest of the SAR algorithm is exactly the same. Assumption is that LOAD is asserted just before CLK going high, and P[7:0] is stable.

[8]

(c) The design of the FSM in Verilog will depend on students. The most straight forward approach is to create 18 states, 16 for the 16 possible outputs, and 1 for waiting for LOAD to go high and 1 for asserting DONE signal. Alternative one could have only 6 states for the six different steps.

[10]

3. This question tests student's ability to map a reasonably complex digital circuit to Verilog, and able to predict the behaviour of the circuit.

(a)

```
module fred (clk, a, z);

    input               clk;
    input               a;
    output reg [2:0]    z;

    reg [1:0]           k;
    reg [2:0]           q, y;

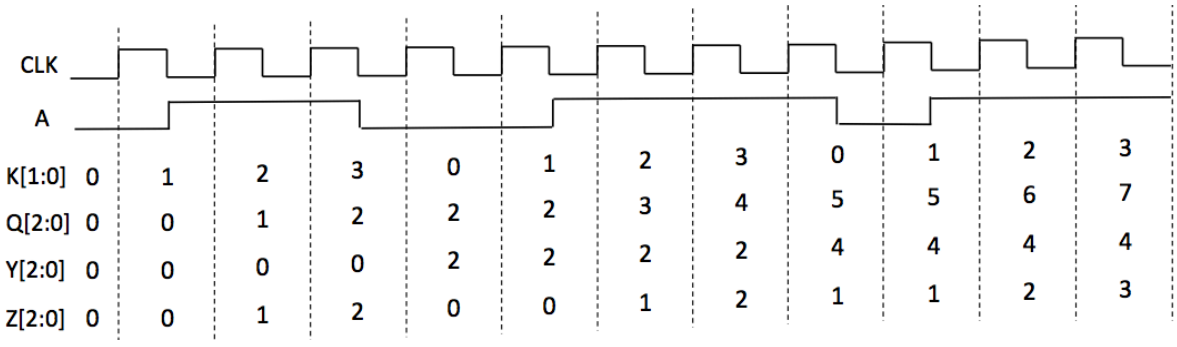
    initial k = 2'b0;
    initial q = 3'b0;
    initial y = 3'b0;

    always @ (posedge clk) begin
        if (k == 2'b11) y = q;
        k <= k + 2'b1;
        if (a == 1'b1) q <= q + 1'b1;
    end     // always

    always @ (*)
        z = q - y;
endmodule
```

[10]

(b)



[15]